System Administration and Maintenance

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4. Implementation of Computer Network

* Planning
* Design
* Implementation.

Learning Outcomes

By the end of this lecture, students should be able to:

- Identify and explain the major components of a computer network.
- Analyze an existing organizational requirement and convert it into network requirements.
- Plan a network using users, applications, traffic, security and growth requirements.
- Design a suitable logical and physical network architecture.
- Select appropriate network devices, media, IP addressing and services.
- Implement, configure, test and document a small computer network.

Roadmap

1. Network requirements analysis
2. Network planning principles
3. Network design process
4. Network components and technologies
5. IP addressing and VLAN planning
6. Physical and logical design
7. Implementation and configuration
8. Testing, troubleshooting and documentation
9. Practical case study and student activity

What Is Network Planning?

- Network planning is the process of determining what the network must achieve before devices are purchased or configured.

- Planning connects business / Organizational requirements to technical requirements.

- Typical planning questions:
  - Who will use the network?
  - What applications and services are required?
  - How many devices/users must be supported?
  - What performance, security and availability are expected?
  - How will the network grow in the future?

Requirement Analysis: Start With the Existing Situation

- Create an inventory of the current environment:
  - Users, departments and locations
  - Computers, printers, servers, phones, cameras and IoT devices
  - Existing switches, routers, access points and cabling
  - Internet connection and service-provider details
  - Current IP addressing and network services
  - Known performance, security and reliability problems

Functional and Non-Functional Requirements

- Functional requirements describe what the network must provide.
  - Internet access, file sharing, printing, email, web applications, VoIP, Wi-Fi
  - Access to servers and shared resources

- Non-functional requirements describe how well the network must operate.
  - Performance and latency
  - Availability and reliability
  - Security and privacy
  - Scalability and maintainability
  - Budget and energy considerations

Network Planning Checklist

- **Users and devices:** How many now? How many in 3–5 years?
- **Applications:** Which applications are business-critical?
- **Bandwidth:** What traffic is expected and when are peak periods?
- **Coverage:** Which rooms, floors and buildings require connectivity?
- **Security:** What users/resources must be separated?
- **Availability:** What happens if a link or device fails?
- **Management:** Who will monitor, maintain and troubleshoot the network?
- **Budget:** What is the total cost of ownership?

Network Components: The Building Blocks

- **End devices:** PCs, laptops, smartphones, printers, IP phones, cameras and IoT devices.
- **Network interface cards (NICs):** connect hosts to the network.
- **Switches:** connect devices within a LAN and forward Ethernet frames.
- **Routers:** connect different IP networks and provide Layer 3 forwarding.
- **Wireless access points:** provide Wi-Fi connectivity.
- **Servers:** provide network services such as DNS, DHCP, file sharing and applications.
- **Transmission media:** copper, fiber and wireless.
- **Security devices/services:** firewalls, IDS/IPS, authentication and access control.

Choosing a Network Topology

- **Star:** each endpoint connects to a central switch; common in modern Ethernet LANs.
- **Extended star:** multiple switches connect in a hierarchical structure.
- **Mesh:** multiple redundant paths; useful where high availability is required.
- **Bus/ring**: historically important, but less common in modern enterprise LANs.
- Selection criteria:
  - Cost and installation complexity
  - Performance and scalability
  - Fault tolerance
  - Ease of troubleshooting

Logical vs Physical Network Design

- Physical design shows where devices, racks, cables and access points are located.
- Logical design shows how traffic is organized:
  - IP subnets and address ranges
  - VLANs and broadcast domains
  - Routing relationships
  - DHCP/DNS and other network services
  - Security zones and access rules
- A strong implementation aligns the physical and logical designs.

Hierarchical Network Design

- **Access layer:** connects end devices such as PCs, printers and phones.
  - Primary functions: endpoint connectivity, VLAN membership and access control
- **Distribution layer:** aggregates access switches and applies policy.
  - Primary functions: inter-VLAN routing, filtering and redundancy
- **Core layer:** high-speed backbone connecting major network segments.
  - Primary functions: fast, resilient transport
- For a small network, access and distribution functions may be combined.

Selecting Network Devices

- **Switch selection factors:**
  - Number and type of ports
  - 1 GbE / 2.5 GbE / 10 GbE uplinks
  - PoE support for APs, IP phones and cameras
  - Managed vs unmanaged operation
  - VLAN, STP, QoS and monitoring support

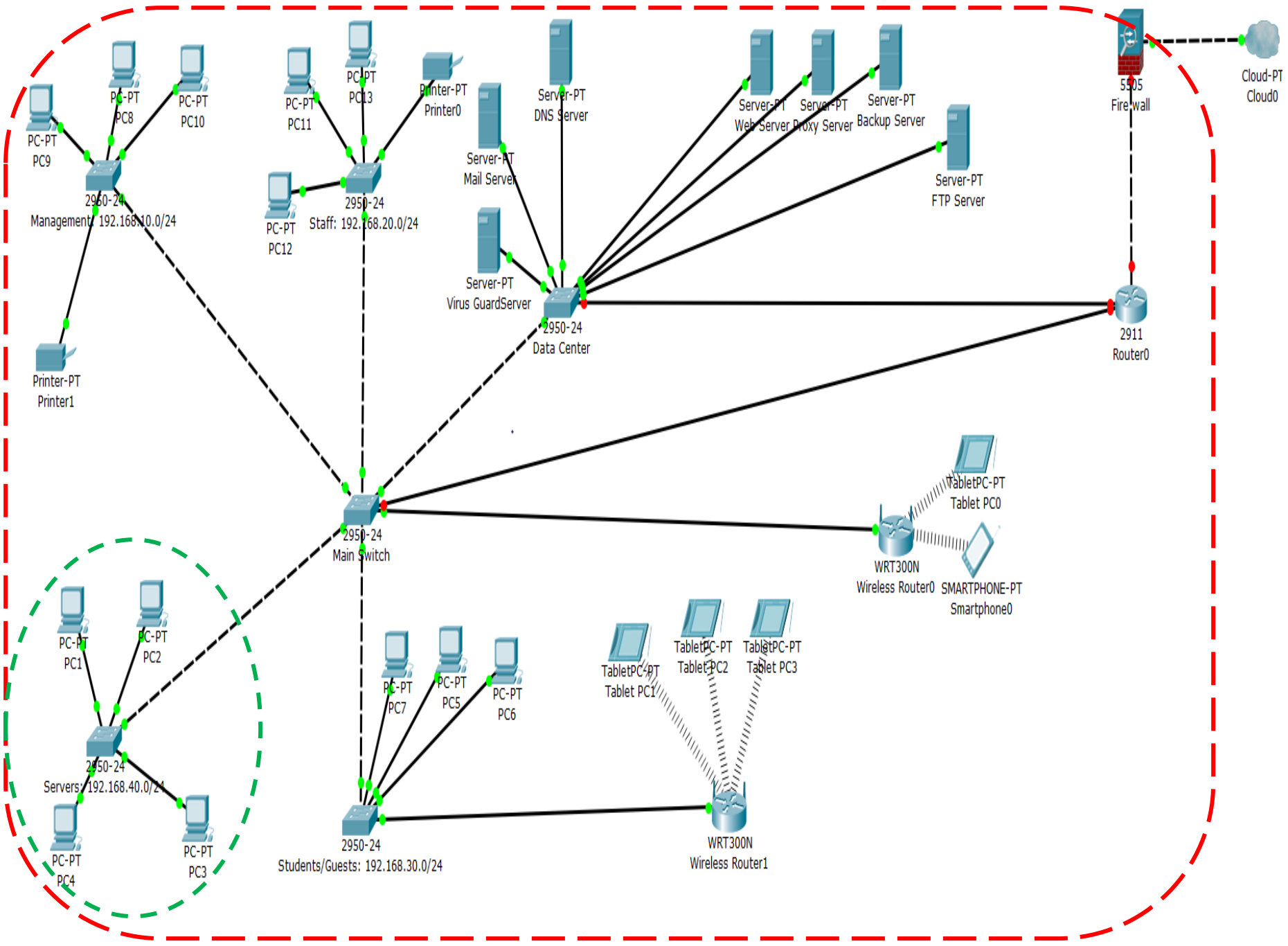
- **Router/firewall selection factors:**
  - Internet speed and VPN requirements
  - Routing and security features
  - Number of users and concurrent sessions
  - Logging, monitoring and management capability

Transmission Media

- **UTP copper:** common for horizontal Ethernet cabling; practical and cost-effective.
- **Fibre optic:** suitable for long distances, high bandwidth and backbone links.
- **Wireless:** provides mobility and flexible deployment but requires RF planning.
- Selection should consider:
  - Distance and bandwidth
  - Electromagnetic interference
  - Installation environment
  - Future capacity
  - Cost and maintenance

IP Addressing Plan

- An IP plan prevents address conflicts and simplifies troubleshooting.
- Example private IPv4 scheme for an organisation:
  - **Management:** 192.168.**10**.0/24
  - **Staff:** 192.168.**20**.0/24
  - **Students/Guests:** 192.168.**30**.0/24
  - **Servers:** 192.168.**40**.0/24
- Document network address, gateway, DHCP range, reserved addresses and DNS.
- Choose subnet sizes based on actual device counts plus reasonable growth.


PC-PT
PC9
PC-PT
PC8
PC-PT
PC10
2960-24
Management: 192.168.10.0/24
Printer-PT
Printer1
PC-PT
PC11
PC-PT
PC13
Printer-PT
Printer0
PC-PT
PC12
2950-24
Staff: 192.168.20.0/24
Server-PT
Mail Server
Server-PT
DNS Server
Server-PT
Web Server
Server-PT
Proxy Server
Server-PT
Backup Server
Server-PT
FTP Server
Server-PT
Virus GuardServer
2950-24
Data Center
5505
Fire wall
Cloud-PT
Cloud0
2911
Router0
2950-24
Main Switch
TabletPC-PT
Tablet PC0
WRT300N
Wireless Router0
SMARTPHONE-PT
Smartphone0
PC-PT
PC1
PC-PT
PC2
2950-24
Servers: 192.168.40.0/24
PC-PT
PC4
PC-PT
PC3
PC-PT
PC7
PC-PT
PC5
PC-PT
PC6
2950-24
Students/Guests: 192.168.30.0/24
TabletPC-PT
Tablet PC1
TabletPC-PT
Tablet PC2
TabletPC-PT
Tablet PC3
WRT300N
Wireless Router1

Subnetting:

- Subnetting divides an IP network into smaller logical networks.
- Benefits:
  - Reduces broadcast scope
  - Improves address management
  - Supports security segmentation
  - Makes routing and troubleshooting easier
- Eg: **/24 provides 256 total addresses.**

  In a typical IPv4 subnet.

  254 are usable for hosts.

VLAN Planning

- VLANs logically separate devices even when they share physical switching infrastructure.
- Eg:
  - VLAN 10 – Administration (10.1.10.0)
  - VLAN 20 – Academic/Staff (10.1.20.0)
  - VLAN 30 – Students (10.1.30.0)
  - VLAN 40 – Servers (10.1.40.0)
  - VLAN 50 – Guest Wi-Fi (10.1.50.0)
- Inter-VLAN communication requires Layer 3 routing.
- Use VLANs with appropriate access-control policies rather than treating segmentation as security by itself.

Network Services Planning

- **DHCP:** automatically provides IP configuration to clients.
- **DNS:** maps names to IP addresses and supports service discovery.
- **Default gateway:** provides access to other IP networks.
- **NTP:** keeps device clocks synchronised for logging and security.
- **File/application servers:** provide shared organisational services.
- **AAA:** supports Authentication, Authorisation and Accounting.
- **Monitoring/logging:** supports performance management and incident investigation.

Security-by-Design

- Security should be considered during planning, not after implementation.

- Key controls:
  - Firewall at the network boundary
  - Strong authentication and least privilege
  - Separate staff, student, guest and management networks
  - Secure Wi-Fi using modern authentication/encryption
  - Disable unused switch ports and services
  - Regular firmware/software updates
  - Backups of configurations and critical data
  - Centralised logging and monitoring

Wireless Network Design

- Plan Wi-Fi from coverage, capacity and security requirements.
- Identify user density and high-traffic areas.
- Select appropriate access-point locations; avoid placing APs solely based on aesthetics.
- Consider channel utilization and interference.
- Separate corporate and guest access.
- Use appropriate authentication and encryption.
- Validate the design with a site survey or post-installation survey.

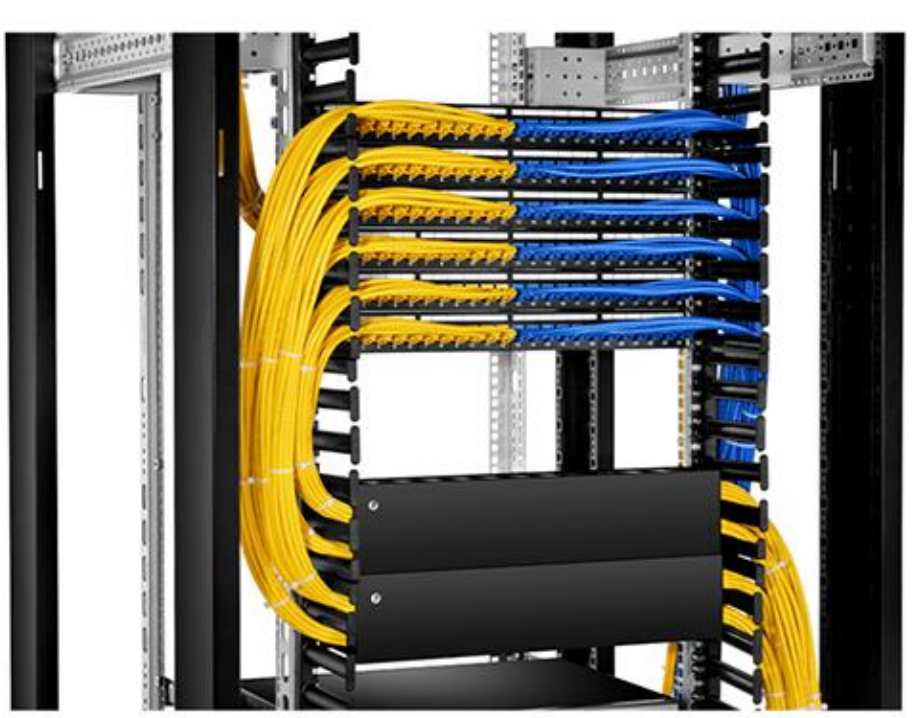
Physical Network Design

- Prepare a floor plan showing:
  - Network rack and patch panels
  - Switches, router/firewall and UPS
  - Cable routes and outlets
  - Wireless access points
  - Server/network equipment locations

- Use clear cable labels and a consistent naming scheme.
- Provide adequate power, ventilation, grounding and physical security.
- Maintain a rack elevation and port-to-outlet mapping.


Network Rack Components Explained
Router
WAN / INTERNET CONNECTION
Firewall
SECURITY PROTECTION
Network Switch
LAN TRAFFIC CONTROL
Patch Panel
CABLE TERMINATION
Servers
NETWORK SERVICES
UPS
POWER BACKUP
PDU
POWER DISTRIBUTION
Cooling & Cable Management
INTERNET
Router
Firewall
Patch Panel
End Devices
(PCs / Printers / APs)
Power Flow:
UTILITY POWER
UPS
PDU
ALL DEVICES
COMPLETE NETWORK FLOW
INTERNET → ROUTER → FIREWALL → SWITCH → PATCH PANEL → END DEVICES


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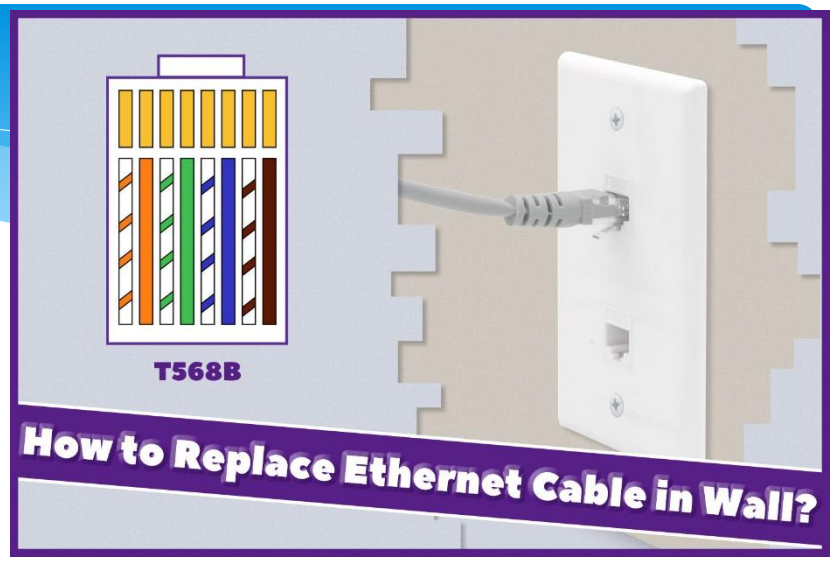
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Mount on Patch Panel

- Installs on the rear of patch panels
- Does not occupy any rack units









https://www.fs.com/blog/8-easy-steps-to-make-ethernet-cable-successfully-4047.html

Logical Network Design Example

- Example: small campus/office
- Internet → Firewall/Router → Core/Distribution Switch → Access Switches
  - Access switches connect PCs, printers, IP phones and APs.
  - Server VLAN contains DHCP/DNS/application services.
  - Guest VLAN has Internet access but no access to internal resources.
  - Management VLAN is restricted to authorized administrators.

Implementation Process

1. Validate requirements and approve the design.
2. Procure and inventory equipment.
3. Prepare rack, power, cabling and labels.
4. Configure devices using a standard baseline.
5. Configure VLANs, trunks, IP addressing and routing.
6. Configure DHCP, DNS, Wi-Fi and security policies.
7. Connect endpoints and apply client configuration.
8. Test connectivity, performance and security.
9. Record final configurations and hand over documentation.

Basic Switch Configuration Concepts

- Set a hostname and management IP address.
- Configure VLANs and assign access ports.
- Configure trunk links between switches where required.
- Enable appropriate loop-prevention mechanisms such as STP.
- Secure management access using SSH rather than insecure remote protocols.
- Disable unused interfaces where appropriate.
- Save and back up the device configuration.
- Exact commands depend on the vendor and operating system.

Router / Firewall Implementation Concepts

- Configure WAN/Internet connectivity.
- Configure internal interfaces or routed VLANs.
- Define routing behavior and default route.
- Configure NAT where required for private-address Internet access.
- Apply firewall rules using least privilege.
- Configure VPNs where remote access or site-to-site connectivity is required.
- Enable logging, time synchronization and secure administrative access.

Testing and Validation

- Test from the bottom of the stack upward:
  - **Physical:** link lights, cable continuity and power
  - **Data link:** VLAN membership, switching and MAC learning
  - **Network:** IP address, gateway and routing
  - **Transport/application:** DNS, DHCP, web, file and application services

- Use controlled test cases and record expected vs actual results.
- Test failure scenarios such as unplugged cables, unavailable services and incorrect addressing.

Troubleshooting Method

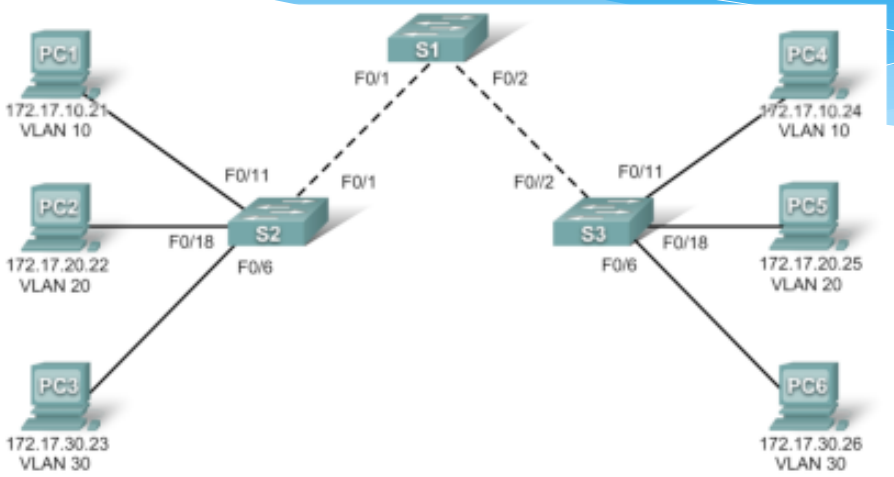
- Use a systematic process rather than changing multiple things at once.
    1. Identify the problem and collect evidence.
    2. Establish a theory of probable cause.
    3. Test the theory.
    4. Plan and implement the fix.
    5. Verify full functionality.
    6. Document the solution.
- Useful evidence: IP configuration, ping/traceroute results, switch port status, logs, DNS results and interface counters.

Network Documentation

- Good documentation is part of implementation—not an optional extra.
- Maintain:
  - Network topology diagram
  - IP address and subnet table
  - VLAN table
  - Device inventory and serial/asset information
  - Switch port and patch-panel mapping
  - Wi-Fi AP locations and configuration
  - Firewall/security rule documentation
  - Backup copies of configurations
  - Change and maintenance records

Basic VLAN Configuration

Topology Diagram



Addressing Table

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 99	172.17.99.11	255.255.255.0	N/A
S2	VLAN 99	172.17.99.12	255.255.255.0	N/A
S3	VLAN 99	172.17.99.13	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	172.17.30.23	255.255.255.0	172.17.30.1
PC4	NIC	172.17.10.24	255.255.255.0	172.17.10.1
PC5	NIC	172.17.20.25	255.255.255.0	172.17.20.1
PC6	NIC	172.17.30.26	255.255.255.0	172.17.30.1

Initial Port Assignments (Switches 2 and 3)

Ports	Assignment	Network
Fa0/1 – 0/5	802.1q Trunks (Native VLAN 99)	172.17.99.0 /24
Fa0/6 – 0/10	VLAN 30 – Guest (Default)	172.17.30.0 /24
Fa0/11 – 0/17	VLAN 10 – Faculty/Staff	172.17.10.0 /24
Fa0/18 – 0/24	VLAN 20 – Students	172.17.20.0 /24

Case Study: KDU ICT Laboratory

- **Requirement:** design a network for a computer laboratory with 40 PCs, 1 lecturer PC, 2 network printers, 2 access points and Internet access.

- **Additional requirements:**
  - Separate student and staff traffic.
  - Provide secure guest Wi-Fi.
  - Centralize IP address allocation.
  - Allow access to required learning services.
  - Support future expansion to 60 PCs.

- **Student task:** propose topology, devices, IP plan, VLANs, cabling, security controls, implementation steps and test plan.

Case Study: Suggested Design

- Internet → Firewall/Router → Managed Core/Distribution Switch
  - VLAN 10 – Staff: 192.168.**10**.0/24
  - VLAN 20 – Students: 192.168.**20**.0/24
  - VLAN 30 – Guest Wi-Fi: 192.168.**30**.0/24
  - VLAN 40 – Management/Infrastructure: 192.168.**40**.0/24

- Two managed access switches can support the laboratory with spare ports.
- APs should broadcast separate staff/student/guest SSIDs as required.
- Apply firewall and inter-VLAN rules based on least privilege.

Practical Activity

- Form groups of 3–4 students.
- **Scenario:** a 50-user office has PCs, laptops, printers, IP phones and Wi-Fi users.
- **Produce:**
  - A network requirement list
  - A physical topology diagram
  - A logical topology diagram
  - IP addressing and VLAN table
  - Device/equipment list
  - Security plan
  - Implementation sequence
  - Testing checklist

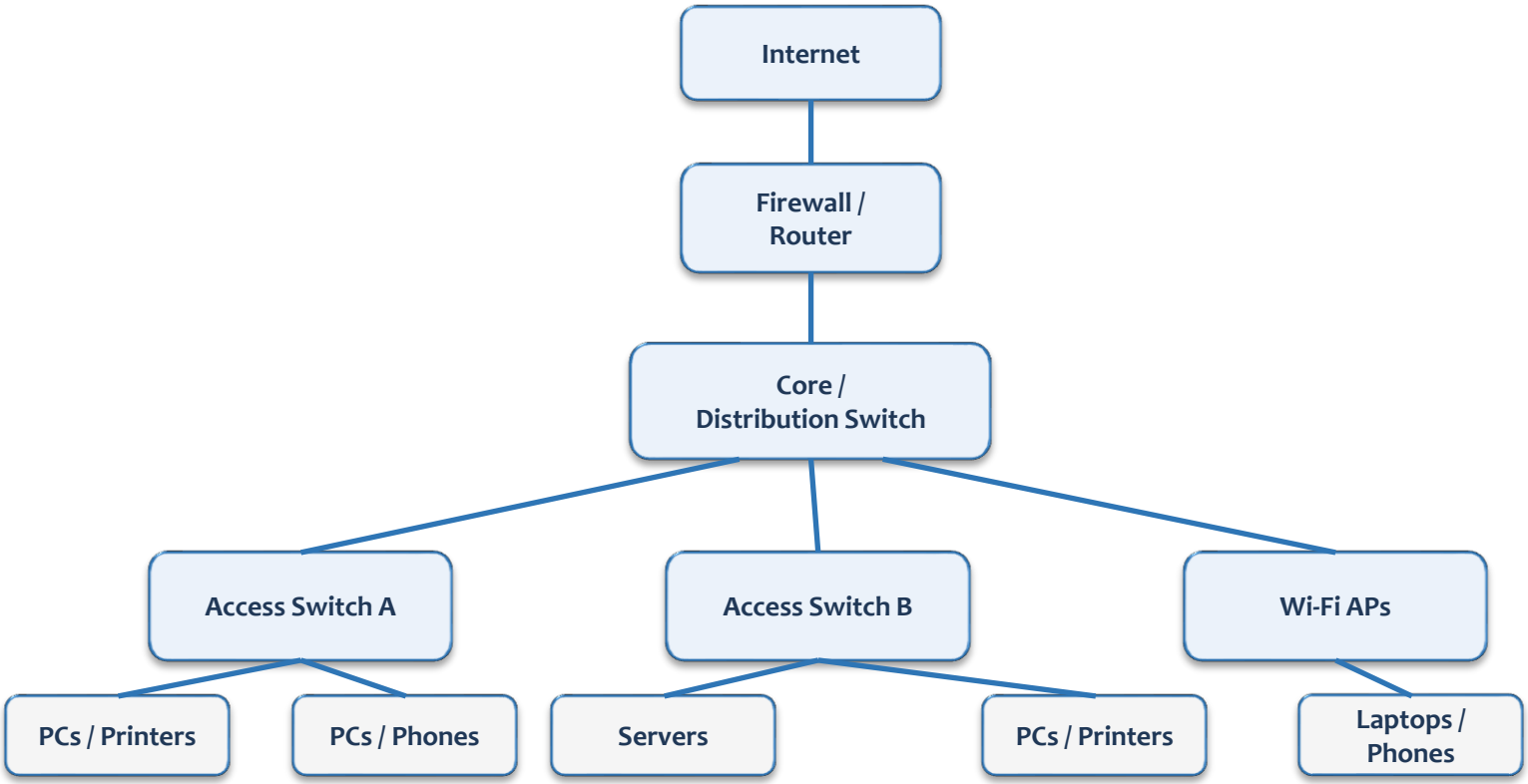
- **Time:** 30–40 minutes; each group presents its design briefly.

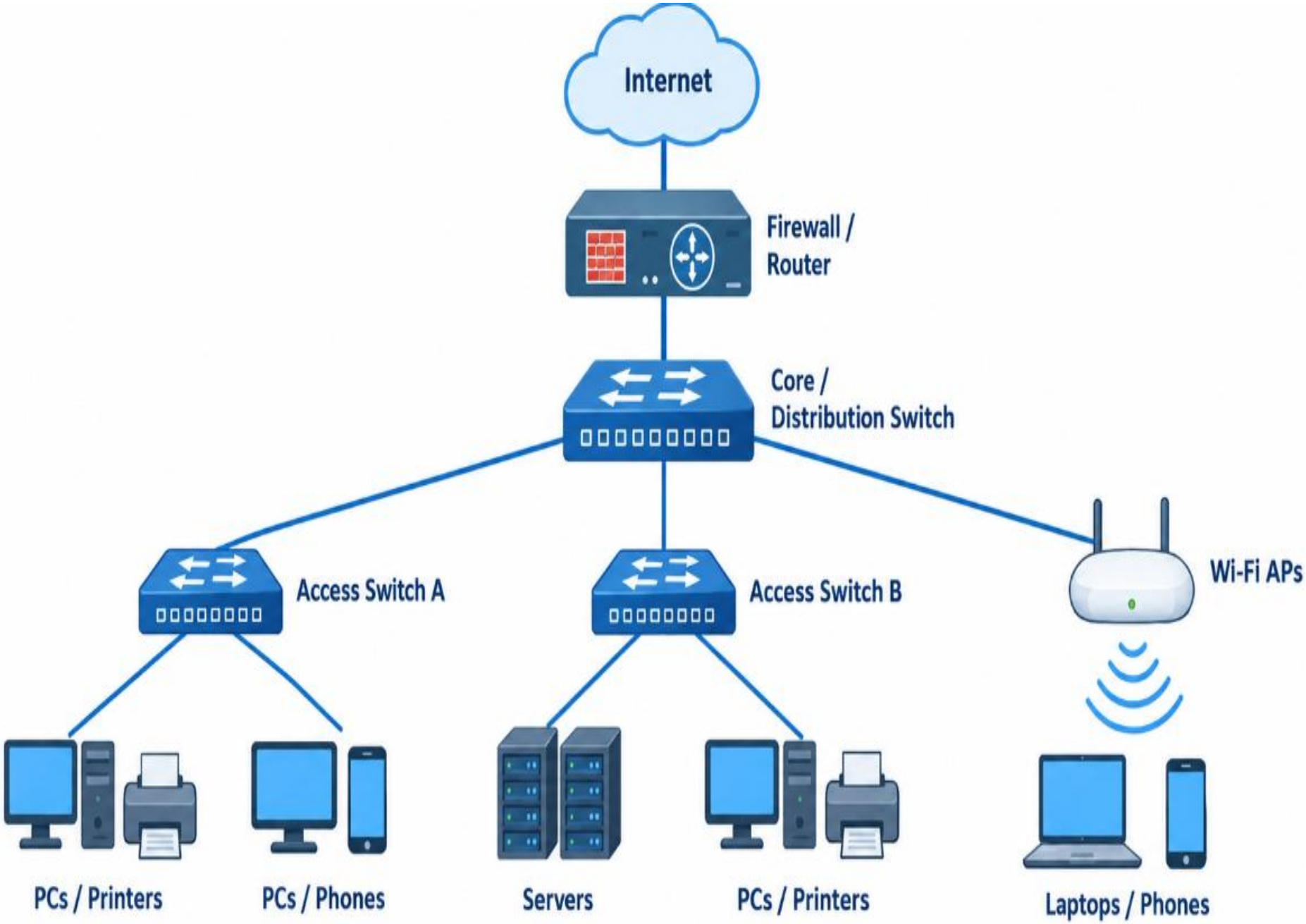
Assessment

1. Why should requirements be analyzed before selecting network devices?
2. Explain the difference between physical and logical network design.
3. When would fibre be preferred over copper?
4. Why are VLANs useful in a university network?
5. What information should an IP addressing plan contain?
6. List five tests you would perform after implementation.
7. What documentation should be handed over after network implementation?
8. How would you design the network to support future growth?

Reference Network Architecture

Example architecture for a small organisation




Internet
Firewall /
Router
Core /
Distribution Switch
Access Switch A
Access Switch B
Wi-Fi APs
PCs / Printers
PCs / Phones
Servers
PCs / Printers
Laptops / Phones

THANK YOU